

HIGH-ENTROPY Li-Na-K FLUORIDE-CHLORIDE MELTS FOR MOLTEN SALT REACTORS

Tkachev N.K.

Institute of Metallurgy UB RAS
620016, Ekaterinburg, Amundsena st., 101

This report presents a theoretical study that evaluates thermodynamic and thermophysical properties of high-entropy molten salt mixtures composed of lithium, sodium, and potassium fluorides and chlorides. These liquid mixtures of six alkali halide components have enhanced thermodynamic stability as a result of the high configurational entropy as expected [1]. The equimolar solution, $\text{Li}_{1/6}\text{Na}_{1/6}\text{K}_{1/6}\text{F}_{1/4}\text{Cl}_{1/4}$, maximizes configurational entropy with the value of $\frac{1}{2}R \ln 24 \approx 1.59R = 13.21 \text{ J/mol}\cdot\text{K}$ approaches the high-entropy threshold of $R \ln 5$ ($13.38 \text{ J/mol}\cdot\text{K}$). Such entropy value significantly exceeds that of FLiNaK ($6.9 \text{ J/mol}\cdot\text{K}$). So they can be promising coolants for molten salt reactors (MSRs) due to improved thermodynamic stability [2]. Two eutectic compositions with reduced lithium content, derived from the phase diagram (melting points 606°C and 630°C), are also analyzed [3]. For example, composition on the phase diagram (point B) in which the proportion is LiF (10%) LiCl (10%) NaF (20%) NaCl (20%) KF (20%) KCl (20%) has high entropy as well: $\frac{\Delta S_{\text{conf}}}{R} = \frac{11}{10} \ln 2 + \frac{1}{2} \ln 5 \approx 1.57$.

Density, heat capacity, thermal conductivity, and viscosity are calculated from liquidus to 1000°C using empirical superposition-extrapolation method, thermodynamic perturbation theory (TPT), as well as molecular dynamics with machine learning interatomic potentials (MLIP).

The diversity of high-entropy coolants offers significant flexibility for tailoring compositions to specific MSR needs, such as fast-spectrum actinide burning or thermal-spectrum efficiency. Through such efforts, high-entropy molten salts could enable MSRs to achieve a closed nuclear fuel cycle, reduce long-lived radioactive waste, and deliver sustainable, carbon-free energy, advancing the goals of Generation IV nuclear technology.

1. Tkachev, N. Three compositions of high-entropy melts of lithium, sodium, and potassium fluorides and chlorides as promising coolants for molten salt reactors. ChemRxiv 2024. <https://doi.org/10.26434/chemrxiv-2024-b6j96>

2. Zakiryanov, D. O.; Tkachev, N. K. High-entropy melt of lithium, sodium, and potassium fluorides and chlorides as a possible coolant for molten-salt reactors. Radiochemistry 2024, 66 (6), 805–809. <https://doi.org/10.1134/S1066362224060093>

3. Zakiryanov, D. O.; Tkachev, N. K., High-Entropy Li-Na-K Fluoride-Chloride Melts for MSRs, J. Phys. Chem. B, 2025, 129 (47), 12276-12284 <https://doi.org/10.1021/acs.jpcc.5c05076>